

Experiment Record

Constellation Diagrams of BFSK, QPSK & 16-PSK and BER vs SNR Analysis

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1. Aim

To generate and plot the constellation diagrams of (a) Binary Frequency Shift Keying (BFSK), (b) Quadrature Phase Shift Keying (QPSK), and (c) 16-ary Phase Shift Keying (16-PSK) at a Signal-to-Noise Ratio (SNR) of 15 dB, both without noise (ideal case) and with Additive White Gaussian Noise (AWGN), and to plot the Bit Error Rate (BER) versus SNR characteristic for all three modulation schemes over an SNR range of -20 dB to 20 dB.

2. Software Used

- Python 3 (with NumPy and Matplotlib libraries)
- Any text editor / IDE capable of running Python (e.g., Jupyter Notebook, VS Code, Spyder) — equivalently, MATLAB may also be used.

3. Theory

3.1 Binary Frequency Shift Keying (BFSK)

BFSK is a digital modulation technique in which the frequency of the carrier is switched between two discrete values, f_1 and f_2 , to represent binary '1' and binary '0' respectively, while the amplitude remains constant. Since the two frequencies are chosen to be orthogonal over a symbol period, BFSK is conventionally represented on a two-dimensional constellation using two orthogonal basis functions (rather than sine/cosine phase axes as in PSK). On this orthogonal I-Q plane, bit '0' maps to the point (1, 0) and bit '1' maps to the point (0, 1), each with unit signal energy, and the two points are separated by 90° , i.e. by a Euclidean distance of $\sqrt{2}$.

3.2 Quadrature Phase Shift Keying (QPSK)

QPSK is a phase modulation scheme that conveys two bits per transmitted symbol by shifting the carrier phase to one of four equally spaced values (45° , 135° , 225° , 315°). Each symbol is Gray-coded so that adjacent constellation points differ by only one bit, minimizing bit errors when a symbol is mistaken for an adjacent one. The four constellation points lie on a unit circle at $(\pm 1/\sqrt{2}, \pm 1/\sqrt{2})$, giving QPSK the same energy-per-bit efficiency as BPSK while doubling the data rate for the same bandwidth.

3.3 16-ary Phase Shift Keying (16-PSK)

16-PSK is a higher-order phase modulation scheme that conveys four bits per symbol by mapping data onto one of 16 equally spaced phase points (spaced 22.5° apart) on the unit circle. While 16-PSK achieves a much higher spectral efficiency (bits/symbol) than BFSK or QPSK, its constellation points are packed much closer together for the same average signal energy. This smaller minimum Euclidean distance between adjacent symbols makes 16-

PSK considerably more susceptible to noise, resulting in a much higher error rate than QPSK or BFSK for the same SNR.

3.4 Additive White Gaussian Noise (AWGN) and SNR

A practical channel corrupts the transmitted signal with AWGN — a zero-mean, flat (white) power spectral density random process with a Gaussian amplitude distribution, added independently to the in-phase and quadrature components of the signal. The Signal-to-Noise Ratio (SNR) is defined as the ratio of signal power to noise power:

$$\text{Desired SNR} = \text{Signal Power} / \text{Noise Power}$$

For a target SNR expressed in dB, the required total noise power to be added to a complex ($I + jQ$) signal of known power is calculated as:

$$\text{Noise Power} = \text{Signal Power} / 10^{(\text{SNR}_{\text{dB}} / 10)}$$

This required noise power is split equally between the in-phase and quadrature noise components (each with variance = Noise Power / 2), and independent zero-mean Gaussian samples of this variance are generated and added to the I and Q components of the modulated signal respectively to simulate the desired channel condition. Since BFSK, QPSK and 16-PSK are all designed here with unit average signal power, the required noise power for the same target SNR of 15 dB works out to be identical for all three schemes, as verified in the Observations section.

3.5 Bit / Symbol Error Rate

The error rate is defined as the fraction of transmitted bits (or symbols) that are received in error, and is the key performance metric used to compare digital modulation schemes. As SNR decreases, noise increasingly corrupts the received symbols, causing the receiver to make more decision errors. For a given SNR, modulation schemes whose constellation points are farther apart (larger minimum Euclidean distance) achieve a lower error rate. Consequently, BFSK and QPSK, whose constellation points are widely spaced, perform considerably better than 16-PSK (16 closely-spaced points) at the same SNR, at the cost of a lower data rate per symbol.

4. Source Code

The following Python program generates random binary data, modulates it using BFSK, QPSK and 16-PSK, adds complex AWGN noise as per the desired SNR, plots the constellation diagrams at SNR = 15 dB (with and without noise), and plots the BER vs SNR curve over the range -20 dB to 20 dB for all three schemes.

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)
N_sym = 5000
N_bits_ber = 200000
snr_const = 15
snr_range = np.arange(-20, 21, 2)
def noise(signal, snr):
    power = np.mean(np.abs(signal)**2)
    n_power = power / 10**(snr/10)
    n = np.sqrt(n_power/2) * (
        np.random.randn(len(signal)) + 1j*np.random.randn(len(signal))
    )
    return signal + n, n_power
# BFSK
bits = np.random.randint(0, 2, N_sym)
bfsk = np.where(bits == 0, 1+0j, 0+1j)
# QPSK
bits = np.random.randint(0, 2, 2*N_sym)
qpsk = ((1-2*bits[0::2]) + 1j*(1-2*bits[1::2])) / np.sqrt(2)
# 16-PSK
M = 16
idx = np.random.randint(0, M, N_sym)
psk16 = np.exp(1j*2*np.pi*idx/M)
# Constellation
for name, signal, lim in [
    ("BFSK", bfsk, [-2, 2]),
    ("QPSK", qpsk, [-1.7, 1.7]),
    ("16-PSK", psk16, [-1.7, 1.7])
]:
    noisy, _ = noise(signal, snr_const)
    fig, ax = plt.subplots(1, 2, figsize=(10, 4))
    ax[0].scatter(signal.real[:800], signal.imag[:800], s=12)
    ax[0].set_title(name + " Without Noise")
    ax[1].scatter(noisy.real[:800], noisy.imag[:800], s=10, alpha=0.5)
    ax[1].set_title(name + " With Noise")
    for a in ax:
        a.set_xlabel("In-Phase")
        a.set_ylabel("Quadrature")
        a.set_xlim(lim)
        a.set_ylim(lim)
        a.grid()
    plt.tight_layout()
    plt.show()
# BER
def ber_bfsk():
    result = []
    for snr in snr_range:
        b = np.random.randint(0, 2, N_bits_ber)
        tx = np.where(b == 0, 1+0j, 0+1j)
        rx, _ = noise(tx, snr)
        b_hat = (rx.imag > rx.real).astype(int)
```

```

        result.append(np.mean(b_hat != b))
    return np.maximum(result, 1e-6)
def ber_qpsk():
    result = []
    for snr in snr_range:
        b = np.random.randint(0, 2, N_bits_ber)
        tx = ((1-2*b[0::2]) + 1j*(1-2*b[1::2])) / np.sqrt(2)
        rx, _ = noise(tx, snr)
        b_hat = np.zeros(N_bits_ber)
        b_hat[0::2] = (rx.real < 0)
        b_hat[1::2] = (rx.imag < 0)
        result.append(np.mean(b_hat != b))
    return np.maximum(result, 1e-6)
def ber_16psk():
    result = []
    k = 4
    for snr in snr_range:
        idx = np.random.randint(0, M, N_bits_ber//k)
        tx = np.exp(1j*2*np.pi*idx/M)
        rx, _ = noise(tx, snr)
        angle = np.mod(np.angle(rx), 2*np.pi)
        idx_hat = np.round(angle/(2*np.pi/M)).astype(int) % M
        result.append(np.mean(idx_hat != idx))
    return np.maximum(result, 1e-6)
b1 = ber_bfsk()
b2 = ber_qpsk()
b3 = ber_16psk()
plt.figure(figsize=(8, 5))
plt.semilogy(snr_range, b1, 'o-', label="BFSK")
plt.semilogy(snr_range, b2, 's-', label="QPSK")
plt.semilogy(snr_range, b3, '^-', label="16-PSK")
plt.xlabel("SNR (dB)")
plt.ylabel("BER")
plt.title("BER vs SNR")
plt.grid(True, which="both")
plt.legend()
plt.show()

```

5. Procedure

1. Generate a stream of random binary data bits using a uniform random number generator.
2. Modulate the bit stream using BFSK mapping on an orthogonal I-Q plane: bit '0' $\rightarrow$ (1, 0), bit '1' $\rightarrow$ (0, 1).
3. Group bits in pairs and modulate using QPSK mapping onto one of four unit-energy phase points at $\pm 45^\circ$, $\pm 135^\circ$.
4. Generate random symbols and modulate using 16-PSK mapping onto one of sixteen unit-energy, equally spaced phase points on the unit circle.
5. Plot the ideal (noise-free) constellation diagrams of BFSK, QPSK and 16-PSK on the I-Q plane.
6. Calculate the average signal power of each modulated signal.
7. For the target SNR of 15 dB, calculate the required total noise power using $\text{Noise Power} = \text{Signal Power} / 10^{(\text{SNR_dB}/10)}$, and split it equally between the I and Q noise components.
8. Generate independent zero-mean Gaussian noise samples of the calculated variance for I and Q, and add them to the respective modulated signals.
9. Plot the resulting noisy constellation diagrams of BFSK, QPSK and 16-PSK at SNR = 15 dB.
10. Repeat the modulation, noise-addition and detection process for SNR values ranging from -20 dB to 20 dB in steps of 2 dB.
11. At each SNR value, demodulate the received symbols (correlator decision for BFSK, sign detection for QPSK, nearest-phase detection for 16-PSK), compare with the transmitted bits/symbols, and compute the error rate for each scheme.
12. Plot the error-rate vs SNR curve (semi-log scale) for BFSK, QPSK and 16-PSK on the same graph for comparison, and record the observations.

6. Observations

6.1 Signal Power and Required Noise Power (SNR = 15 dB)

Modulation	Signal Power	Desired SNR	Required Noise Power
BFSK	1.0000	15 dB	0.03162
QPSK	1.0000	15 dB	0.03162
16-PSK	1.0000	15 dB	0.03162

6.2 Constellation Diagram - BFSK (SNR = 15 dB)

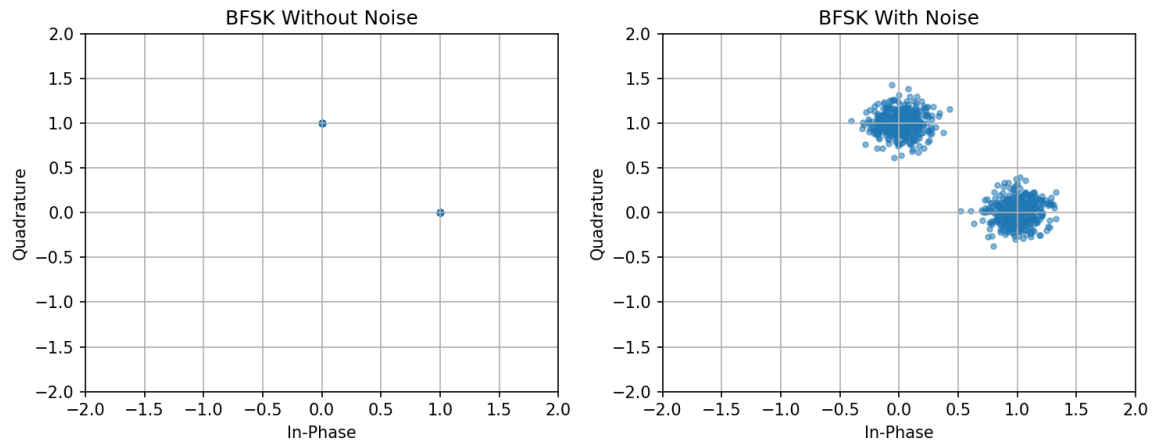


Fig. 1: BFSK constellation without noise (left) and with AWGN at SNR = 15 dB (right).

6.3 Constellation Diagram - QPSK (SNR = 15 dB)

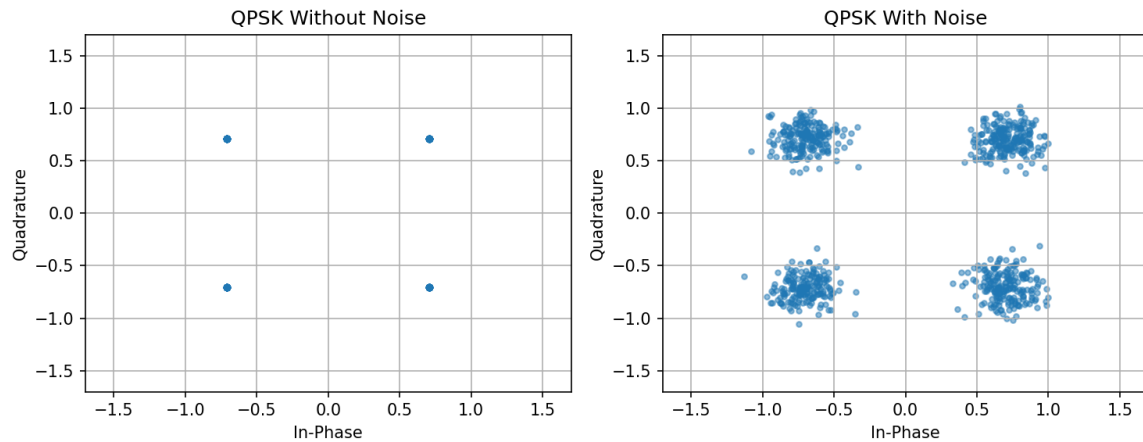


Fig. 2: QPSK constellation without noise (left) and with AWGN at SNR = 15 dB (right).

6.4 Constellation Diagram - 16-PSK (SNR = 15 dB)

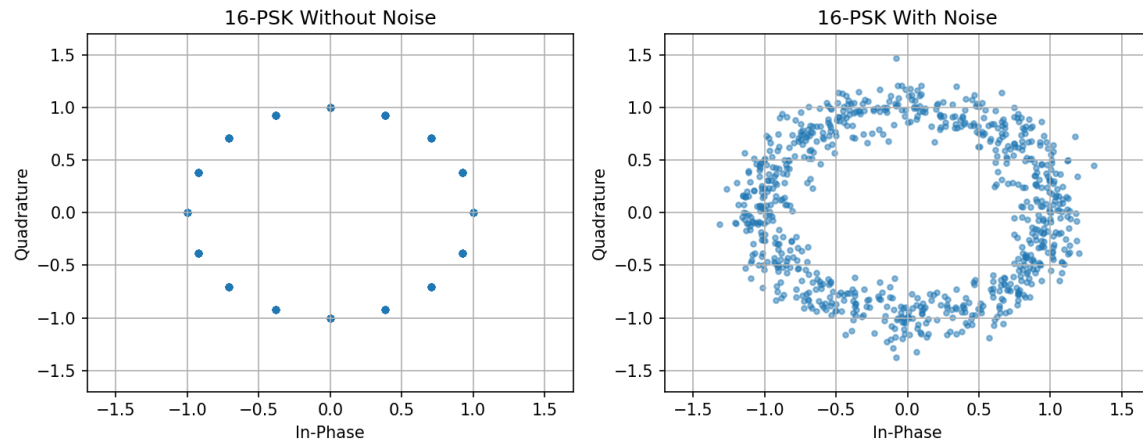


Fig. 3: 16-PSK constellation without noise (left) and with AWGN at SNR = 15 dB (right).

6.5 Error Rate vs SNR (-20 dB to 20 dB)

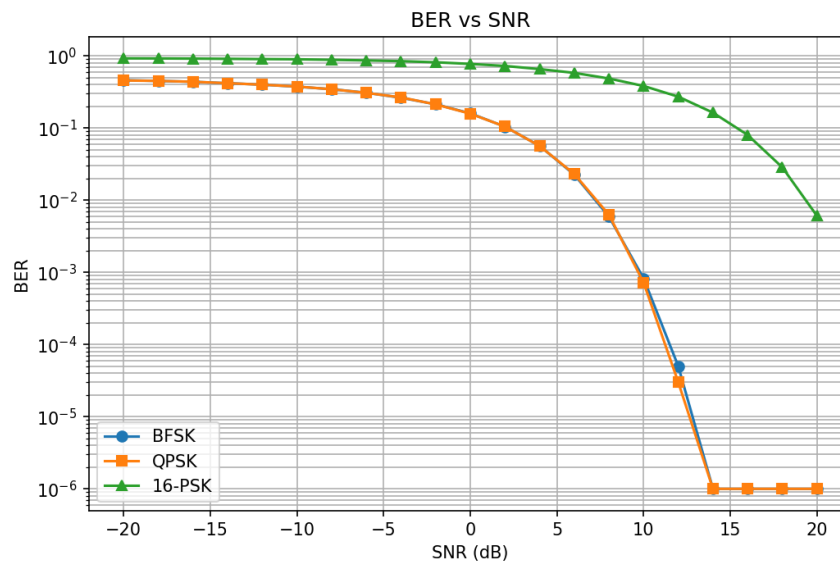


Fig. 4: Error rate versus SNR for BFSK, QPSK and 16-PSK.

6.6 Sample Readings

SNR (dB)	-20	-10	0	8	12	16	20
BFSK	0.4613	0.3746	0.1599	0.00593	0.00005	~0	~0
QPSK	0.4577	0.3767	0.1584	0.00628	0.00003	~0	~0
16-PSK	0.9269	0.8990	0.7754	0.4866	0.2726	0.0804	0.0061

7. Results & Discussion

The constellation diagrams confirm the theoretical expectations for all three schemes. Without noise, BFSK shows two sharp points on orthogonal axes, QPSK shows four sharp points at $\pm 45^\circ$ offsets, and 16-PSK shows sixteen sharp, equally-spaced points on the unit circle. When AWGN is added at SNR = 15 dB, each constellation point spreads into a cluster whose radius is governed by the common noise variance (0.0316, identical for all three schemes since each has unit average signal power). For BFSK and QPSK, whose constellation points are widely separated, these clusters remain clearly distinguishable at 15 dB. For 16-PSK, however, the sixteen closely-packed points visibly blur into a near-continuous ring, since adjacent points are only 22.5° apart.

The error-rate vs SNR plot shows that BFSK and QPSK track each other very closely across the entire range and both fall off sharply, reaching a near-zero error rate by about 14 dB, since both schemes have a comparatively large minimum distance between constellation points relative to their signal energy. 16-PSK, in contrast, exhibits a markedly higher error rate at every SNR value and requires a much larger SNR to approach a low error rate, because packing 16 points onto the same unit circle drastically reduces the minimum distance between adjacent symbols, making 16-PSK far more vulnerable to phase perturbation by noise for the same signal power. At very low (negative) SNR, all three schemes tend toward a high error rate since detection becomes increasingly unreliable as noise dominates the signal.

This result illustrates the fundamental trade-off in digital modulation: higher-order schemes such as 16-PSK pack more bits per symbol (greater spectral efficiency) but sacrifice noise immunity, while lower-order schemes such as BFSK and QPSK are more robust to noise but carry less information per symbol.

8. Conclusion

The constellation diagrams of BFSK, QPSK and 16-PSK were successfully plotted, both without noise and with AWGN at SNR = 15 dB, and the effect of noise in spreading the constellation points was clearly observed, most prominently for 16-PSK due to its closely packed symbol points. The error rate vs SNR characteristic was plotted for the range -20 dB to 20 dB, confirming that the error rate decreases as SNR increases for all three schemes. It was further concluded that BFSK and QPSK offer very similar, and considerably better, noise performance than 16-PSK, which trades noise immunity for a higher data rate per symbol — demonstrating the classic power-versus-bandwidth-efficiency trade-off among digital modulation schemes.